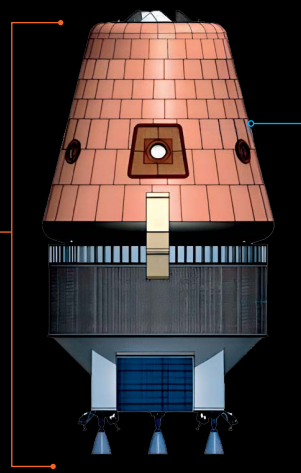
KEY SYSTEMS

HUMAN-RATED LVM3

ISRO's heavy lift launch vehicle will launch the orbital module, including crew module (CM), to low Earth orbit of 400 km. It will also have a crew escape system to ensure that the CM along with the crew is taken to a safe distance in case of an emergency during lift-off



ORBITAL MODULE

Will orbit Earth and comprises the CM and Service Module (SM). It is equipped with a state-of-the-art avionics system with adequate redundancy, considering human safety. SM will be used for providing necessary support to CM while in orbit

CREW MODULE

It is the habitable space with Earth-like environment in space. It houses the crew interfaces, human-centric products, life support system, avionics and deceleration systems. It is also designed for re-entry to ensure safety of the crew during descent

Graphic by TANMOY CHAKRABORTY

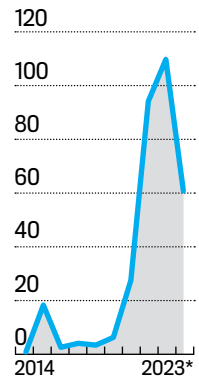
INDIA'S NEW SPACETECH ECOSYSTEM

SKY IS THE LIMIT

Propelled by privatisation since 2020, the Indian space technology sector is on an upward trajectory

7th
India's rank in terms of funding within the international spacetech ecosystem in 2023

INVESTMENT TRENDS



*(In $mn; *till July)*
Source: Tracxn

TOP 3 COMPANIES*

Pixxel	97.1
Skyroot	72.3
AgniKul	34.8

**Based on total funding*

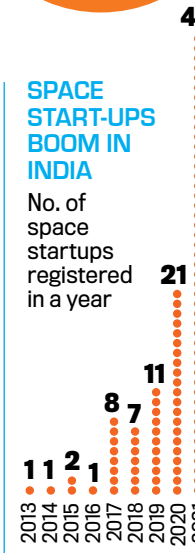
TOP 5 FUNDING DEALS

Skyroot	51
Pixxel	35
MTAR	24.4
AgniKul	20
SatSure	14.5

(Since 2020; in $mn)

SPACE START-UPS BOOM IN INDIA

No. of space startups registered in a year



Source: Startupindia

and use ISRO's extensive facilities, including launch pads and laboratories, to do so. A coordinating government body called the Indian National Space Promotion and Authorisation Centre (IN-SPACe) chaired by Pawan Goenka, ex-CEO of Mahindra & Mahindra, was formed for private players to interface with ISRO and use its facilities. In April this year, the Union government formalised the process by announcing the Indian Space Policy 2023 to institutionalise private sector participation.

Already, the policy is yielding results with the new private space tech ecosystem for commercial rocket launchers and satellites expanding at a meteoric pace. Over 150 space start-ups have been set up and many, like Skyroot, Pixxel and AgniKul, have received substantial foreign funding. In November 2022, Skyroot became the first Indian private sector company to indigenously build a rocket launcher and successfully launch it into sub-orbital space from Sriharikota. The company has received an impressive $51 million (Rs 421 crore) in investment since 2020 and a total of $72.3 million (Rs 597 crore) since its inception five years ago. Overall funding for space start-ups has grown exponentially, attracting $112 million (Rs 925 crore) in 2022. Funding to private entities has already touched $62 million (Rs 512 crore) this year, a 60 per cent rise over the same period last year. Says Jitendra Singh, the Union minister of state for science & technology and the departments of space and atomic energy, "All this is because the prime minister took the courageous decision to completely open up the space sector to private entities in the country. For too long, we had unnecessarily put a veil of secrecy over our space activities and confined ourselves to ISRO. This was no longer required. As a result, there is a quantum jump in our space missions." (See interview, "*India is no longer the lesser cousin*".)

Private investment and initiatives will be important if the space sector in India has to expand rapidly, because it cannot depend on government resources alone to fund projects. As Singh points out, "If we have to visualise a global role for ourselves, we need global parameters and a global strategy. That is what the US government did and NASA, its space agency, no longer depends on government sources for funds or expertise to execute missions." Of course, it will take Indian industry a while to reach where, say, Elon Musk has with SpaceX and Jeff Bezos with Blue Origin in the US. According to Somanath, the satellite and launch market constitutes only 30 per cent of the world's $360 billion (Rs 29.7 lakh crore) space economy; it is space applications that account for 70 per cent of the business. And this is where Indian private companies can make a major foray. India's current share of the space business is around $8 billion (Rs 66,000 crore) or just over two per cent of the total. By 2040, if all goes well, per global consultancy firm Arthur D. Little, India could realise its potential of being a $40-100 billion space industry and become a major global player. That is a mission India should pursue at rocket speed. ■